

1 Title

0:00

SAY: *"Good morning. My thesis is a design study on visual analytics for EEG artefact removal — a tool called ICARUS, built with EEG experts here at TU Graz."*

cue — reset timer · breathe · look up

2 Design Study Methodology

~1:00

► *"First, how I approached this."*

- A **design study** is **problem-driven** visualisation research: work with real experts on a real problem and build a tool — opposite of technique-driven work that invents an algorithm in isolation.
- Nine stages, three phases: Precondition (learn, pick collaboration), Core (discover, design, implement, deploy), Analysis (reflect, write) — validated by the experts throughout.
- Key point: the contribution can be a **well-justified combination of existing techniques** — no new algorithm required.
- Ran it with the Institute of Neural Engineering over several months — meetings, prototype reviews, interviews.

Not here — don't enumerate the nine stages.

3 Electroencephalography (EEG)

~1:45

► *"The data we work with is EEG."*

- Measures the voltages from **neurons' post-synaptic potentials** at the scalp, at **millisecond** resolution — great for timing.
- **Non-invasive, portable, low-cost** → a workhorse in research and the clinic.
- Used for epilepsy diagnosis, sleep staging, brain–computer interfaces, cognitive studies.
- Electrodes on the standard **10–10 system**; 32-channel subset here (left: layout, right: real cap).
- Raw EEG is essentially **unusable as-is** — needs preprocessing; conclusion quality depends on signal quality.

Not here — the artefacts (next slide).

► *"And every recording is full of non-brain signal."*

- An **artefact** = any potential not from the brain; can be **larger than the brain activity itself** → dominates the trace (fig: blinks burying the frontal channels).
- From the body (eyes, muscle, heart) and from outside (power line, movement).
- They **overlap the brain signal in time and frequency** → can't just band-pass or threshold them away.
- Slip through → corrupt the analysis (wrong artefacts in, wrong conclusions out).

Not here — the specific types (next slide).

5 Artefact Types

~3:15

► *"Two families — and each behaves differently."*

Physiological (from the body)

- **Ocular** — eye movements/blinks; eyes sit next to the brain → **most common & impactful**; large, slow, low-frequency, frontal; measured by EOG.
- **Cardiac** — the heartbeat; **rhythmic sharp spikes** mirroring the ECG, strongest over the left temporal area.
- **Muscle (EMG)** — head/neck contractions; random, non-repetitive; **below 1 Hz to over 200 Hz** → hits every band; on all sensors → the **hardest** to remove.

Technical (external)

- **Line noise** — 50 Hz mains, constant on all channels; removed by a notch filter.
- **Electrode/channel** — poor contact/dry gel → jumps, flat lines, a single noisy electrode.
- **Movement** — cable/electrode shifts → sudden deflections. **Environmental** — nearby devices.
- Punchline: **line noise** → **preprocessing**; physiological ones overlap the brain and must be **separated** → **ICA**. Muscle is the hard one.

Not here — how we separate them (ICA, next).

6 Independent Component Analysis

~4:15

► *"The standard way to separate them is ICA."*

- Unmixes the electrode mixture into **statistically independent components**, one per source.
- **Cocktail-party problem**: microphones each pick up a mixture of voices; ICA recovers the voices. Electrodes = microphones.
- $\mathbf{X} = \mathbf{A} \cdot \mathbf{S}$; estimate $\mathbf{W} \approx \mathbf{A}^{-1}$ so $\hat{\mathbf{S}} = \mathbf{W} \cdot \mathbf{X}$; works by maximising non-Gaussianity (FastICA).
- Each component carries a **topography, a spectrum, and a time course** → these become the three views.
- $\# \text{components} \leq \# \text{electrodes}$; default **16** per recording.

Not here — classification (next slide).

7 Manual Component Classification

~5:00

► *"But ICA doesn't tell you which component is brain and which is artefact."*

- An **expert looks at every component** and decides keep (brain) / reject (artefact) from the three properties.
- Rejected ones removed by **back-projection** — signal rebuilt from kept components (fig: blinks red → gone in black).
- **Irreversible & asymmetric:** wrong reject deletes real signal; wrong keep leaves noise → keep the human in control.
- **The bottleneck:** hundreds of participants $\times$ ~ 16 components → slow, rushed, inconsistent; reasoning never recorded.

cue — fig is MNE sample data, illustrative.

8 Existing Tools

~6:00

► *"Don't existing tools already do this?"*

- **MNE-Python** — richest primitives; `plot_properties` (topo+spectrum+epochs) + per-artefact detectors, but **no unified classifier, no calibrated confidence**.
- **FieldTrip** — MATLAB; ICA classification is script-driven/manual; no automated step, no cross-recording view.
- **EEGLAB** — GUI grid + per-component popups; cross-subject clustering needs manual setup; classifier = ICLabel plugin.
- **ICLabel / MARA / ADJUST** — automated labels/probabilities, but no interactive interface, fixed outputs.
- Table: **no single tool combines** linked views + cross-participant overview + batch-apply + calibrated confidence.

Not here — don't narrate every cell; don't say "novel" — own the combination.

9 Research Questions

~6:45

► *"This gives me two research questions."*

- **RQ1** — which visualisation & interaction techniques support expert decision-making in ICA artefact classification?
- **RQ2** — how can ML recommendations and interactive visualisation be integrated to accelerate classification while preserving expert control?
- These two are the contract for the rest of the talk.

cue — read both slowly · PAUSE.

► ***"Six requirements came straight out of the design study."***

- Derived from the **discovery interviews** — each is a real need, not a guess.
- **Overview & guidance** — R1 keep/reject recommendations · R2 an overview of all components.
- **Component inspection** — R3 over time · R4 brain vs artefact · R5 scalp distribution.
- **Workflow** — R6 log & export decisions for reproducibility and team consistency.
- Every feature maps back to one of these.

Not here — don't read all six word-for-word — group them.

► *"Briefly, how it's built."*

- **Two Docker containers:** Flask backend (preprocessing, ML inference, files) + React/nginx frontend, REST API.
- Standard stack: MNE-Python, scikit-learn, aeon/MiniRocket, umap-learn; loads XDF.
- Frontend draws every visualisation on **HTML canvas**.
- Defaults (50/100 Hz notch, 1–75 Hz, 250 Hz, FastICA 16) tuned to the lab, **configurable per study**.

Not here — don't dwell, it's plumbing; ML pipeline is backup.

12 Classification Workflow

~9:00

► *"The whole workflow lives in one tool — end to end."*

- **(A) Preprocess** — create a study (multi-participant, shared config), upload, run.
- **(B) Apply recommendations** — model scores every component; apply the confident ones in bulk.
- **(C) Inspect** — open the coordinated views for the uncertain ones.
- **(D) Control clusters** — check the cross-participant overview for outliers; then **export** + optional retrain.

Not here — don't explain each view (own slides next).

13 Confidence Recommender

~10:00

► *"First building block — the recommender."*

- A **keep-probability** per component (1 = brain, 0 = artefact).
- Three tiers (five bands, symmetric around 50%): **Tier 1** $\geq 75\%$ / $\leq 25\%$ (batch-apply), **Tier 2** 60–75% / 25–40% (quick review), **Tier 3** 40–60% (manual).
- Principle: **classification with a reject option** — uncertain cases go to the expert.
- Muscle is hard → a **separate badge** from MNE's detector (faded 50–70%, solid >70%).
- **Indicator only** — never pre-sets the decision; expert confirms each (came from the evaluation).

Not here — ML internals/accuracy = backup. ~40% in Tier 1 only if asked.

14 Participants & Components Overview

~10:45

► *"You start from this overview."*

- **Left:** every participant with their tally — kept / rejected / undecided of total.
- **Right:** that participant's components, each with a confidence bar, a one-click tier button, and a muscle badge.
- One **Apply**: auto-rejects undecided muscle-flagged components, then applies Tier 1 to the rest — **never overwrites manual labels** (reset available).
- Click a row → opens the inspector.

Not here — the inspector detail (next slide).

15 Component Inspector

~11:30

► *"Click a component and the inspector opens — the heart of the tool."*

- All views for **one component on a single screen**: recommender bar, cluster, topomap, spectrum, timeline.
- Move to another component (row, cluster point, arrow keys) → it stays open and **every linked view updates together**.
- That's what the **script workflow can't do** — there you regenerate static plots per component and lay them out yourself.
- Quote: "the entire visualisation on one screen, more or less."

DEMO LOOP — *narrate it; look at the committee, not the screen.*

Not here — don't explain each view; topomap / PSD / timeline get their own slides.

16 Component Cluster & Mapper Graph

~12:15

► *"And a study-wide overview across all participants."*

- Both project **every component from every participant** into one shared 2D space (UMAP on MiniRocket descriptors) — similar components sit close.
- **Cluster**: one dot per component, coloured by label (or by confidence).
- **Mapper**: nearby components grouped into pie-chart nodes (sized by count) + edges — structure without overplotting.
- Best use = **outlier detection**: a dot whose colour differs from its cluster flags a likely misclassification → overview first, drill down.

Not here — the actual caught case is the evaluation; don't over-claim.

► ***"First detail view — topography, the expert's first criterion."***

- How strongly each sensor contributes + its polarity, from the ICA mixing matrix (μV); interpolated; blue $\rightarrow$ white $\rightarrow$ red.
- The patterns classify: **dipolar = brain · frontal = ocular · peripheral patch = muscle · single electrode = bad channel.**
- Hover = electrode name + amplitude; re-computes for a selected time range.

18 Power Spectral Density

~13:15

► *"Second detail view — the spectrum, the other primary criterion."*

- Power across frequency, the five EEG bands shaded as background.
- The shape names the source: **falling 1/f (often an alpha peak) = brain** · **flat/rising = muscle** · **delta/theta = ocular**.
- Updates to whatever time range you select on the timeline.

► ***"Third detail view — the timeline, used to confirm."***

- The component's signal over the whole recording; scroll = time zoom, drag = pan, y-axis = amplitude.
- **Right-click-drag selects a range** → drives the topomap + PSD to that segment.
- Three overlays: **EOG markers** (blink events; if spikes align → ocular), **trial markers** (is the artefact even inside a trial?), **PSD clusters**.
- Role is confirmatory — eye artefacts = the classic large, slow deflections.

Not here — the PSD-cluster detail (next slide).

► *"One overlay is worth a closer look — PSD clustering."*

- **K-means on the spectra of overlapping time windows** across the participant's components → colour-codes the timeline by spectral cluster.
- Configurable cluster count, window size, step; similar spectral profiles get the same colour.
- Payoff: **see at a glance where a component's frequency character changes**; hover a segment for the cluster's mean spectrum.

21 Evaluation

~15:15

► *"I evaluated ICARUS with two domain experts on their own EEG data."*

- **Think-aloud** + semi-structured interview, on **real BCI recordings** from their lab — the data and task they do day to day.
- **Five hands-on tasks**: recommendations · artefact identification · band-power analysis · mapper exploration · view comparison.
- Consistent **decision sequence** (index → topomap → spectrum → timeline) matched the layout. Recommender trusted on ocular; muscle checked by hand.
- **Key moment**: the cross-participant overview let the expert **catch a mislabeled component** — imperfect classifier + expert inspection corrects errors the model alone would miss.

CLOSE (no slide): *"So the value is the integration — the recommender handles the clear cases, the expert keeps control of the rest. Thank you, I'm happy to take*

your questions." (PAUSE before "Thank you".)
B Backups (Q&A only)

Not here — don't volunteer limitations; keep them for Q&A.

- **Recommender Pipeline** — each component's time series → MiniRocket (9,996 feats) + 16 unmixing-matrix weights → Random Forest → keep-probability → tiers; ~75% test on 480 components / 30 participants. Lead with retrainability; don't volunteer "second-best model"; ICLabel → answer verbally.
- **Classification Experiments** — the nine-experiment table; MiniRocket chosen for speed (Matrix Profile 79% vs 75% but needs a self-join).
- **Data Flow** — decisions export as YAML, re-import into MNE (`ica.exclude` → `ica.apply()`), metadata preserved; labels can retrain.

Q&A — listen → rephrase → pause → answer briefly. Press **O** for the overview to jump to a backup.